EDHEC BUSINESS SCHOOL

MSc in Financial Engineering

27_M2_NI_FENG_S1_CCO_FIN_2764

ADVANCED FIXED INCOME AND CREDIT

GROUP ASSIGNMENT

Submission date: **November 1st 2026**

Lecturer: Thibault Lair

INSTRUCTIONS:

- **Assignment type:** Group assignment
- **Assignment instructions:**
  - Groups should work independently.
  - If something is not clear, please make a reasonable assumption and continue. Please highlight your assumptions so that I can identify them when grading the assignment.
  - In case of questions, channel your questions through a group representative.
- **Submission instructions:**
  - Submit your assignment by email and through Blackboard, by October 31st 2025 midnight (Paris time).
  - You should submit a single pdf or word file containing all your answers to the questions as well as the personal information of all group members (school identifier, first and last name). Different files per questions are not allowed. Please indicate clearly the group number.
  - You are allowed to attach additional files (e.g. Python) to support your answers. Submitting them will allow for grading partially your reasoning in case of erroneous final answer. Please reference them clearly in your main document if you want them to be considered. They should be properly commented to make it easy for me to go through them.
- **Case study:**
  - The assignment is based on a Harvard Business School case study "Deutsche Bank: Finding Relative Value Trades". You can download the case and the associated data by using the below links:
    - Case: https://hbsp.harvard.edu/tu/9967d21b
    - Data: https://hbsp.harvard.edu/tu/cde6b860
- **Python labs:**
  - The assignment is based on a Python Labs Jupyter notebook "US Treasury Yield Curve Analysis: Trump Tariff Threats Impact". You can download from Blackboard the following files available in the 'Evaluation/Group assignment' folder:
    - Jupyter notebook: "pylab_trump_tariff_threats_assignment.ipynb"
    - Python script: "fred.py"

CASE STUDY: Deutsche Bank: Finding Relative Value Trades

QUESTION 1: Bootstrapping

Q1.1:

Solve for the prices of discount bonds given the market prices of coupon bonds and identify the zero-coupon yields[1] from the prices of zero-coupon bonds.

Q1.2:

Compare the zero-coupon yields from Deutsche Bank's model (in Exhibit 4 of the case) to the market zero-coupon yields, and identify which zero-coupon bonds Deutsche Bank should buy, and which ones they should sell.

Q1.3:

Is this strategy (of buying some bonds and selling others) a risk-free arbitrage strategy?

QUESTION 2: Cubic splines

Q2.1:

Estimate the cubic splines' function that uses as knots 2-, 5-,10-, 15- and 20-year zero-coupon yields. Describe succinctly the methodology and document any choice made in the estimation. Plot the discount function (i.e., the cubic splines based ZC yields) against the zero-coupon yields obtained in Q1.1.

Q2.2:

Discuss when using those new 'spline-based' zero-coupon yields estimates instead of the one implied by the bootstrapped zero-coupon bonds in Q1.1 is appropriate. Please justify and discuss whether this affects the conclusions reached in Q1.2?

QUESTION 3: Nelson Siegel model

Q3.1:

Estimate the discount function according to the Nelson-Siegel-Svensson model. Describe succinctly the methodology and document any choice made in the estimation. Plot it against the zero-coupon yields obtained in Q1.1.

Q3.2:

Estimate and plot the factor loadings and contributions from the different factors affecting the shape of the spot zero-coupon curve. Explain their key drivers.

QUESTION 4: Hedging

Q4.1:

To estimate the sensitivity of any instrument or portfolio to the 10-year rate changes, i.e., its 10-year key rate duration (KRD), you will shock the 10-year rate by 50bps. Discuss what the impact is on the zero coupon yield curve estimated in Q2.1? Plot the difference.

Q4.2:

Assume you own a 1 million $ portfolio that is equally invested in all bonds. What is the KRD of your portfolio to this scenario? Describe your approach to estimating this exposure.

Q4.3:

Estimate and plot the KRDs of your portfolio for all maturities, when applying the same shock individually to all zero-coupon yields. Plot and interpret the results.

[1] Be mindful of the yield and compounding convention used in the case study. Adopt a consistent approach throughout the assignment.

PYTHON LABS: Trump tariffs' threats

Your objective is to evaluate the impact of Trump trade war on the US yield curve. To this aim, you will conduct an event study and analyse the abnormal yield curve change around specific event dates.

Follow the instructions and answer the below questions.

INSTRUCTIONS:

Preparation:

- Create a "functions_library" sub-folder in the FIN2764 project and drop the "fred.py" there.
- Drop the 'pylab_trump_tariff_threats_assignment.ipynb' in the 'module_1' folder. Use it as a starting point.

Measurement:

To define the abnormal yield curve change around specific events, follow the below steps:

- For all event dates in the sample (see notebook), measure the yield curve change (relative to the yield on the event date) over a window of [-30,+30] business days.
- Measure the abnormal yield change as the average yield changes over all identified event windows vs the average of all other non-event dates.

QUESTION 1: Impact evaluation

Q1.1:

Investigate how markets have reacted to Trumps announcements/actions. Discuss

Q1.2:

Extend your analysis to the different yield curve components (slope, level, curvature). Discuss

Q1.3:

Same question as Q1.1 when it comes to the reaction of targeted countries. Discuss

Q1.4:

Have the reactions been different in the first vs second mandate? Discuss

QUESTION 2: Methodology

Q2.1:

Please discuss the potential limitations of the above analysis. If any, propose methodological improvements and evaluate how they affect your responses in question 1.